

Types of genetic diseases (disorders):

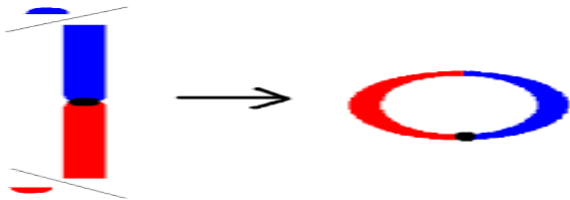
Each human is estimated to have approximately 20,600 different coding genes. Alterations in these genes, or re combinations of them, can produce genetic disorders.

- 1-Chromosomal disorders.
- 2-Single gene disorders.
- 3-Multifactorial disorders.
- 4-Mitochondrial disorders.

- **Chromosome abnormalities (anomalies) aberrations**
- Certain genetic diseases are caused by alterations in the number or structure of chromosomes. Usually because of (mistakes) during meiosis or mitosis, homologous chromosomes or their chromatids may not separate (non disjunction), the result is an abnormal number of chromosomes in the gamete produced and later, in the zygote.
- Numerical anomalies result in either **polyploidy**, the addition of complete haploid sets of chromosomes or **aneuploidy**, the addition or loss of one, two or rarely more chromosomes.

Structural anomalies result from breakage and fusion of chromosome segments in a novel way (deletion and translocation, etc.). A deletion is simply the loss of a portion of a chromosome. A translocation occurs when a deleted portion of one chromosome is transferred to and becomes part of another, non homologous chromosome.

*Ring chromosome(both ends of a chromosome are fused).



*Isochromosome(Both chromosome arms are a mirror image to each other,in which a chromosome is consisting of two copies of either short arms or long arms)

The **polyploidy** conditions that have been observed in humans are triploidy (69.XXX) and tetraploidy (92, XXXX). The additional chromosomes encode a large amount of gene product, causing multiple anomalies such as defects of the heart and CNS. Triploidy and tetraploidy are both lethal. The most common cause of triploidy is the fertilization of an egg by two sperms (dispermy). Triploidy can also be caused by the fusion of an ovum and a polar body. Tetraploidy can be caused by a mitotic failure in the early embryo.

- **Autosomal aneuploidy:**
- Autosomal aneuploid conditions consist primarily of monosomies and trisomies. They are usually caused by non disjunction, the failure of chromosomes to disjoin normally during meiosis I or II. The resulting gamete either lacks a chromosome or has two copies of it, producing a monosomic or trisomic respectively. Autosomal monosomies are always lethal, while some autosomal trisomies are compatible with survival.

Partial autosomal monosomy:

1-Cry du chat syndrome(46xy-5p)

2-Hirschhorn syndrome(46xy-4p)

- **Trisomy 21 (Down syndrome) 47, X X + 21 or 47, X Y +21:**
- Down syndrome, formerly called **Mongolism**. It seen in approximately 1\800 live births.
- Persons with D.S usually have three copies of chromosomes 21 in their karyotype, about 95% of cases are caused by non disjunction during formation of parental gametes, the remainder being caused by chromosome translocation, most often chromosomes 14 and 21 joined together.
- D.S includes characteristics facial appearance (folding of the eyelid a low nasal root, a large tongue and a flattened maxillary region), short stature, heart defects, and an increased susceptibility to infection and mental retardation. The chances of a woman having a Down syndrome child increase rapidly with age, starting at about age forty. Although people with D.S, on average, have a lifespan much shorter than normal, some live to middle age or beyond.

- **Trisomy 18 (47, X Y + 18) Edwards syndrome:**
- This trisomy is the second most common autosomal trisomy, with prevalence of about 1/6,000 live births. Trisomy 18 patients have prenatal growth deficiency, characteristic facial features (include small ears, a small mouth that is often hard to open), a short sternum, hand abnormalities, congenital heart defects (ventricular septal defect), and protrusion of the bowel in to the umbilical cord. About 50% of children with trisomy 18 die in the 1st month, and only about 10% are still alive at 12 months of age.

- **Trisomy 13 (47, X Y + 13) patau syndrome:**
- This condition is seen in about 1\10,000 births. The malformation consists of oral-facial cleft, microphthalmia, polydactyly and abnormal C.N.S the survival rate is very similar to that of trisomy 18, with 90% of affected fetuses are spontaneously aborted. The risk of bearing a child with this condition increases with advanced maternal age.

- **Sex chromosome aneuploidy:**
- Non disjunction may also produce individuals with unusual number of sex chromosomes, but the consequences of this class of aneuploidy are less severer than those of autosomal aneuploidy.
- **Turner syndrome (45, XO):**
- Turner syndrome is the only known viable human monosomy. About one girl in 3,000 has T.S. These individuals are phenotypically female, their sex organs do not be mature at adolescence, and secondary sex characteristics fail to develop. Such individuals are sterile (do not undergo puberty or menstruate).
- Girls with T.S exhibit short stature, have a broad chest and a webbed neck. Many patients with T.S have congenital heart defects. There is typically some diminution in spatial perceptual ability, but females with T.S are not mentally retarded.

- **Klinefelter syndrome (47, XXY):**
- About one young man in 1,000 male births is found to have K.S. These men tend to be taller than average, with long arms and legs, they are not usually mentally retarded. Affected individuals are sterile males, the testes are underdeveloped and there may be some breast development. These phenotypic abnormalities are not apparent until puberty. It is estimated that at least half of 47, XXY conceptions are spontaneously aborted.
- Individuals with the 48, XXXY and 49, XXXXY karyotypes have also been reported because they have a Y chromosome, they still have a male phenotype, but the degree of mental deficiency and physical abnormalities increase with each additional X chromosome.

- **47, XYY syndrome:**
- Males with this karyotype tend to be taller than average, suffer from persistent acne, and tend to have barely normal intelligence. At one time, it was suggested that these men were likely to be criminally aggressive, but it has since been shown that the incidence of such violent behavior among them is no greater than among XY males.

- **Trisomy X (Meta female):**
- The 47, XXX karyotype occurs in 1\1,000 females and usually has benign consequences. In some cases there is a tendency toward physical abnormalities, some times suffer from sterility, menstrual irregularity, or mild mental retardation.
- Females have also been seen with 4, 5 or even more X chromosomes. Each additional X chromosomes accompanied by increased mental retardation and physical abnormalities.

- **XX males and XY females:**

- During normal meiosis in the male, cross over occurs between the tip of the short arm of Y chromosome and the tip of the short arm of the X chromosome. These region of X and Y chromosomes contain highly similar DNA sequences. The distal portion of the Y chromosome is known as the pseudo autosomal region. Occasionally, the cross over occur just outside the pseudo autosomal region, where lies agene known as SRY (sex-determining, region on the Y). this gene which expressed in embryonic development, encodes a product that interacts with other genes to initiate the development of the undifferentiated embryo in to a male. As a result of a faulty crossover between the X and Y chromosomes, such that the SRY gene, instead of remaining on the Y chromosomes, is transferred to the X chromosome. An offspring receiving this X chromosome will be an XX male, and an offspring receiving the Y chromosome that lacks the SRY gene would be an XY female.